

04 Secret Messages

This assignment challenges you to write several small programs to work with cryptographic hashes and ciphers.

Setup

You are welcome to use your language and environment of choice. I recommend working with Python and its cryptography package, which is documented here:

<https://cryptography.io/en/latest/>

If you use your Duke VCM virtual machine, it is pretty easy to get set up. Your Linux VM already has Python installed. You need to update the cryptography package as follows:

```
sudo apt-get install -y python3-pip
pip3 install cryptography --upgrade
```

Thereafter you can edit programs with `nano` and run them with `python3`.

Problem 1

Write a program to compute the SHA-3 512-bit hash of the Howard and Lipner book. You can get the file from the course web site:

http://people.duke.edu/~jk471/fintech514.s21/Microsoft_Press_eBook_TheSecurityDevelopmentLifecycle_PDF.pdf

Format your hash value as hexadecimal digits. Submit both the hash value and your program.

Big old hint: you can verify your answer with `openssl` using a command like this:

```
openssl dgst -sha3-512 TheSecurityDevelopmentLifecycle_PDF.pdf
```

Problem 2

Write a program that uses the supplied key and initialization vector to decrypt the given ciphertext. The encryption was performed using the AES algorithm, with PKCS#7 padding and Cipher Block Chaining (CBC) mode.

```
Key       : aa51df2cffdb99ecf84a0faa7d7b7d1f7f3f995184aaa2b0731ca626e6ce3166
IV        : e236b0f8a07a971ddbb06d41b6fb4867
Ciphertext : 808850d7fec9c691b3775631e827f6032b4cc3088c8b3127e71a469821b18fb1
```

Submit your answer (the plaintext) and the program that you used to find the answer.

Problem 3

Create a key and a nonce. Use them with an AES cipher in CTR mode (not CBC this time!) and PKCS#7 padding to encrypt some plaintext of your choosing. Submit the key, nonce, and ciphertext as hexadecimal, just like I gave you in Problem 2. Also submit the program you used to do the encryption.